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Operational experiences in meteor burst telemetry — eight years of SNOTEL project observations

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Abstract
Communication over long distances using meteor trails has recently been applied to large scale data acquisition projects.

The U.S. Department of Agriculture Soil Conservation Service's SNOTEL meteor burst telemetry system consists of 510 sites located in high mountain areas of the Western United States.

The SNOTEL system consists of two master stations which initiate interrogation and receive data transmissions and forward their data to a central computer system. A typical SNOTEL remote data site is equipped with sensors which provide snowpack, rainfall, and temperature data.

Meteor burst communications are a reliable alternative to other types of telemetry. The SNOTEL system accurately senses and transmits data to a central facility. Data users can conveniently access the system.

Data from the SNOTEL meteor burst system are used operationally in snowmelt runoff forecasting and many other ways. The Soil Conservation Service is reducing its traditional manually measured snow course networks and placing increased reliance on SNOTEL data.

Introduction
Communication over long distances using meteor trails has been used for many years. The technique of using these trails for communication was discovered by ham radio operators who, under certain conditions, found themselves talking to someone halfway around the world. The principle gained widespread attention when radio contact with astronauts was temporarily lost during reentry. A shield of ionized gasses around the space capsule during time of reentry repelled the radio signals.

The U.S. Department of Agriculture Soil Conservation Service initiated a meteor burst telemetry system called SNOTEL in 1977. By late 1980, there were 475 sites operating. Currently, there are 510 sites located in high mountain areas of the Western United States. The SNOTEL system is the world's largest known application of this method of telemetry.

Meteor burst telemetry
Meteor burst technology takes advantage of the billions of meteorites that enter the Earth's atmosphere daily. These meteorites range in size from cosmic dust to rock. As each particle enters the Earth's atmosphere, it burns, producing an ionized trail of gasses in a zone 80 to 120 kilometers (50 to 80 miles) above the Earth's surface. (See figure 1.)

These trails diffuse rapidly, usually disappearing within seconds. During their brief existence, however, they will either reflect or reradiate radio signals, particularly those in the low end of the VHF range of 40 to 100 megahertz (MHz). The height of these trails allows radio communications from a transmitter to be received as far as 1,920 kilometers (1,200 miles) from the source (Leader, 1974) and also either eliminates or substantially reduces the adverse effects of local terrain obstacles.

Meteors--defined as extraterrestrial objects in elliptical orbit around the sun--are divided into two classes: shower and sporadic. Shower

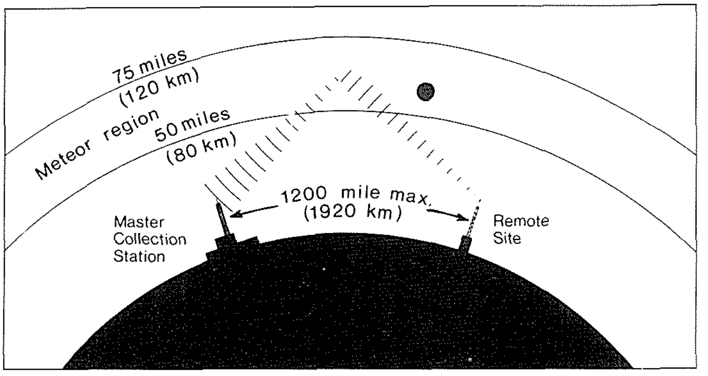

FIG.1 Meteor burst data acquisition system.

meteors are groups of particles all moving with the same velocity in well-defined orbits around the sun. They enter the atmosphere in spectacular fashion and appear to observers on Earth to be radiating from a common point in the sky. Sporadic meteors move in random orbits around the sun. Most meteors used in radio work are in this class. Although their random entry points make them less predictable than shower meteors, certain characteristics are known. Intersection of their orbits with the Earth's orbit is most frequent in August and least frequent in February, at a ratio of about 4:1 due to the tilt of Earth's axis. The forward motion of the Earth sweeps about four times more meteors into the atmosphere in the morning than during the evening, resulting in a significant diurnal variation in available meteor trails. (See figure 2.)

The ionized trail of gasses is composed of positively charged ions and free electrons. The electrons reflect or reradiate the radio waves,

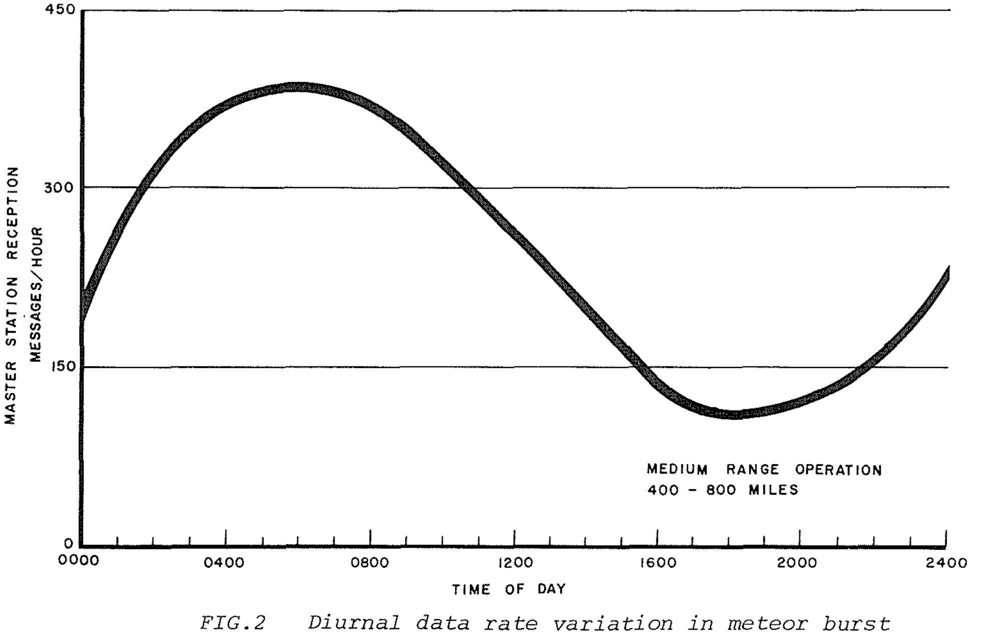

FIG.2 Diurnal data rate variation in meteor burst communication.

depending on the density of the trail. Typical trails are about 25 kilometers (15.5 miles) long with a radius of about 1 meter (40 inches). Low electron line densities (underdense trails) result in reradiation of the radio signal. Overdense trails reflect the radio signal. Signals from underdense trails rise to an initial peak volume in a few hundred microseconds, then typically decay within from a few milliseconds to a few seconds.

Other ways in which the meteor burst principle can be used to communicate include reflection of the radio signals off of objects such as airplanes at distances of less than 160 kilometers (100 miles). These reflections usually last several seconds and are characterized by a great deal of signal fading. The E region, from 100 to 125 km above the Earth's surface, provides long periods of communication time. However, the sporadic nature of the E region limits its usefulness.

Communication over an operating range of up to 1920 km (1200 miles) is possible. Sites in close proximity to each other can be reached via line of site radio signals, or ground wave. Curvature of the Earth and terrain features generally limit this type of communication to about 160 km (100 miles), although ground wave has been observed to extend up to 240 km (150 miles) from the transmitting station to the receiving location. Beyond the ground wave signal, the number of messages per unit of time that can be expected is a function of the amount of sky that can geometrically reflect signals from the master to the remote site. This is referred to as "common sky." The common sky is less at closer distances, increases to its maximum in the range of 800 to 1,280 km (500-800 miles), and then reduces with distances beyond 1,280 km (800 miles). (See figure 3.)

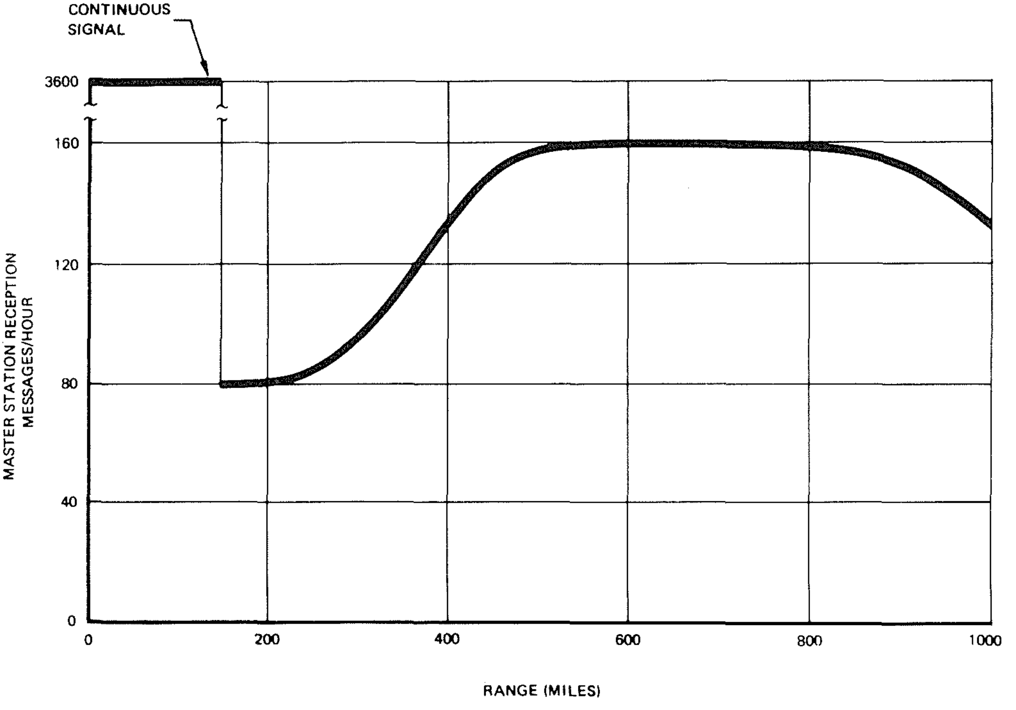

FIG.3 Relationship between meteor burst communication rate and range in distance between master and remote.

It was not until the advent of microprocessor based communication equipment that the meteor burst technique became useful in large scale data transmission projects. Prior to the age of rapid computer processing, requirements of such a system could not be satisfied by manual operators.

SNOTEL

SNOTEL has been in operation in the Western United States since 1977, when the first remote site data were telemetered via very high frequency (VHF) radio using meteor burst technology. This system's primary function is to sense and transmit hydrometerological data from remote mountainous sites to the central data collection station in Portland, Oregon.

By October 1980, 475 sites had been completed and the network was fully operational. Since then, 35 more sites have been added, bringing the total operational sites to 510.

The SNOTEL system is composed of two types of stations: remote data acquisition sites, and master stations--one near Boise, Idaho, and the other near Ogden, Utah. The entire network of remote sites is within communication range of each master station. Using two master stations enhances system performance and substantially reduces communication failures resulting from problems at the master stations.

An optimum SNOTEL remote site is situated in a small clearing in the timbered high mountains. The surrounding forest canopy provides a sheltering of the catchment area. Therefore, snow accumulates uniformly, and precipitation falls in a nearly vertical path for optimum gage catch. The site is also sheltered from winds, thus minimizing transport of the snowpack either onto or away from the measurement area. The site aspect is typically flat to north facing, minimizing solar radiation effects on the snow. The slope is gentle, allowing for free water drainage, but not steep enough to permit snow creep. Stainless steel "pillows," about 3 cm (1.2 inches) thick and 1.8 m^2 (20 ft^2) in surface area, are used to hydraulically sense the weight of the snowpack. Pillows are normally placed in configurations of 3 or 4 units which are connected together to function as a single weighing platform. Precipitation is collected in a 30.5 cm (12 inch) diameter cylinder. The height of the gage is a function of the annual amount of rainfall expected, as well as of the depth of the winter snowpack. Both the water equivalent of the snowpack and the amount of accumulated precipitation are measured by pressure transducers. Air temperature is also sensed at the site. An analog signal is relayed from the transducers, converted to digital format, and forwarded to a storage buffer within the communications unit every 15 minutes. Data are then transmitted to a master station upon receipt of a valid probe.

All sites in the network are interrogated once daily during a 3-hour period from 0500 to 0800, Pacific Standard Time. Additional polls can be conducted on either a scheduled or ad hoc basis, depending on data needs. Newer models of remote site electronics have microprocessors capable of acquiring additional data. These upgraded sites can be programmed to store and report several observations per day, such as total 24-hour wind run, maximum, minimum, and average wind speed and temperature during a 24-hour period, and other predefined functions.

Another recently implemented feature of the new models is the daily transmission of data regarding the operational characteristics of the site. Such diagnostic parameters as signals received, transmissions, master station acknowledgment received, battery voltage under load, and transmit forward and reverse power are now available to field maintenance technicians. Using this information, the technician can detect operational problems and perform repairs prior to failure of the site.

The SNOTEL system was designed to provide maximum control and flexibility to the operational offices. Data are collected at the master stations in response to either scheduled or unscheduled polls. On a set fifteen minute cycle, all data are forwarded to a central computer facility in Portland, Oregon. The data are filed and kept on line and available for up to 14 months. This allows access to the current water year's (October 1 through September 30) data plus two months. The five Soil Conservation Service offices charged with responsibility for remote site operations and data quality are able to readily access the central system via remote computer terminal. They examine the "raw" data, make editing adjustments, and perform analyses. Other agencies and organizations which require SNOTEL data for their operations may also gain access via terminal.

At the conclusion of the current water year, the data are subjected to a final examination, and editing is performed to assure that the final record diverges no more than ± 10 percent from the onsite "true" value. When this process is completed, the data are placed in an archival data base at the U.S. Department of Agriculture's computer facility in Fort Collins, Colorado.

Data Reliability

The performance of the SNOTEL system is evaluated based on three criteria: The first criterion relates to the consistency with which data are received from all sites during the regularly scheduled, or nominal poll. The goal is to acquire data from at least 90 percent of the sites. See table 1 for a summary of nominal poll performance. Hardware and software failure accounts for virtually all performance problems, with very little failure attributable to lack of adequate meteor bursts. The reader will note a marked improvement in system performance during mid-1985. This improvement is attributed to two factors. First is continued improvements in field servicing. Both the quality of service equipment and the skill of the field technicians continues to improve. Second is improvement in software designed to conduct master station to central computer communications and to process the data. The inevitable bad data files, which must be anticipated to occur occasionally, are now being processed using improved data recovery techiques, thus salvaging some data reports that previously would have been lost and treated statistically as non-reporting sites.

Table 1. Systemwide response to nominal polls
Monthly average percentage of sites reporting

	Daily nominal polls				
	1981	1982	1983	1984	1985
January	--	81.6	89.3	79.2	88.4
February	74.1	78.8	88.2	82.5	87.5
March	80.2	83.8	91.1	85.4	91.5
April	82.5	85.6	90.0	87.5	93.4
May	82.7	90.4	89.5	90.5	96.1
June	80.4	89.2	91.6	90.8	96.5
July	81.7	88.8	90.9	89.4	
August	84.1	86.6	89.4	90.1	
September	--	87.7	88.9	89.9	
October	85.7	89.7	90.2	89.3	
November	86.0	88.7	86.5	90.7	
December	85.5	87.2	80.6	89.5	

The second criterion is the diurnal stability of data. Sensors and electronic communication equipment are maintained to keep diurnal change at, or less than, 0.3 percent of full scale capacity of the sensor, i.e., 7.6 mm (0.3 inch) in a 2,540-mm (100 inch) device. A majority of the data channels meet this standard of acceptability, as table 2 illustrates. The targeted minimum percentage of data channels meeting this criterion is 80 percent. The sensitivity of electronic and mechanical components to temperature changes is illustrated by the more stable diurnal fluctuations in the winter months when snowpacks insulate equipment and daily temperature ranges are small. During the summer when equipment is exposed to large daily temperature ranges, data channels fluctuate more widely.

Table 2. Diurnal fluctuation of data channels

	Percentage of channels with 0.3 percent or less diurnal fluctuation				
Month	1981	1982	1983	1984	1985
February	80.4	85.4	90.9	84.7	90.4
June	80.0	67.6	74.1	75.0	73.5
October	80.4	75.6		92.0	

The third criterion sets the required accuracy of sensors and data transmission. Data stored at the central computer facility should be within 10 percent of actual onsite conditions. Conditions are measured periodically by survey crews and these measurements are compared with telemetered data. If SNOTEL data are outside the $\pm$ 10 percent limits, those data are edited and brought into conformance with the observed values. The predominance of data errors appear to be caused by system component malfunctions, with a lesser percent attributable to sensor problems.

Snow pressure pillows tend to systematically overweigh the accumulated snowpack, as reported by Smith and Boyne (1981). The magnitude of measurement error is dependent primarily on snowpack physical properties. Ice lenses can cause temporary over or under weight, lasting from a few hours to a few days. A well located, properly installed pillow system should be expected to yield data which are no more than about 5 percent greater than actual or true values as determined by excavation and weighing of the snowpack on the pillow surface. This systematic overweight is generally equivalent to the systematic bias in the commonly used Federal snow sampler. Thus, measurements by both techniques should be equal (Work, et. al, 1965).

In a critical test of 5 years of SNOTEL performance, Schaefer and Shafer (1982) studied telemetry and sensor performance in the Colorado-New Mexico part of the system. They concluded that there was no appreciable systematic error, that is, observed and telemetered values were equivalent. In their study random error resulted in an average expected difference of about $\pm$ 1.25 centimeters (0.5 inch) between observed and telemetered data. They compared data gathered by snow survey teams using standard Federal snow samplers to telemetered snow-water data and found a nearly 1:1 correspondence in arrays of four metal pillows and about a 5 percent systematic overweigh in three-pillow systems.

Using SNOTEL data in streamflow forecasting

SNOTEL data are retrieved by computer terminal for a wide variety of uses. The principal use by the Soil Conservation Service is in forecasting snowmelt season volumes, and hydrograph recession on snowmelt streams. These forecasts are generated primarily for irrigation water management and reservoir operations.

Much of the justification for the existence of the SNOTEL system and the resultant streamflow forecast is based on recurrent drought in the Western United States. The most recent severe drought occurred in 1977. Streamflow forecasts were widely used to optimize the use of all the region's available water. The severity of any drought needs to be judged objectively so that mitigating water management actions can be taken. The recently developed Surface Water Supply Index (SWSI), described by Shafer and Dezman (1982), is a method of evaluating the severity of water supply problems, using SNOTEL data, streamflow forecasts, and reservoir storage data.

SNOTEL data are used operationally in streamflow forecasting by SCS in three ways. The first use is in physical process modelling of streamflow. SNOTEL derived snowpack, precipitation, and air temperature data are being input as parameters in hydrologic basin models. Selected forecast points are currently being calibrated and tested. Output from these models includes peak flow, time of peak, recession characteristics, low flow, seasonal volume, and other hydrograph features. More widespread use of these models is anticipated in 1986 as additional forecast points are calibrated.

Secondly, SNOTEL data are being used to improve the accuracy of the snowpack variable in regression forecasting procedures. Traditional regression procedures have used the first of the month snow course data as one of the most significant independent variables. A problem arises when the maximum seasonal accumulation of snow occurs at the snow course between the scheduled measurements. This situation is presumed to occur commonly between April 1 and May 1, but prior to the availability of daily SNOTEL data, could not be confirmed and quantified. Now with SNOTEL, the forecasting hydrologist can make the needed adjustment to the first of the month measurement, thus improving the resultant forecast reliability. A variation of this theme is the revision of snowpack data based on SNOTEL observations and recalculation of forecast values when major short duration storms are detected over the watersheds catchment area.

The third way that SNOTEL data are being used operationally is in substituting SNOTEL data for manually observed snow course data. This procedure requires a correlation of SNOTEL and snow course data. A regression equation is then used to estimate snow course values from SNOTEL reports. These estimates are used in lieu of measured snow course data in the existing regression procedures to forecast streamflow volumes. This method is being used extensively by the SCS pending the accumulation of sufficient data history that SNOTEL data can be used directly in multiple regression analyses.

Conclusions

Although meteor burst communications has been a useful technique for many years, it was not until the advent of high speed computer processing that the technology became a viable data telemetry option. In 1977, the SCS began installation of what became the world's largest known meteor burst telemetry system--a project called SNOTEL.

Eight years experience with SNOTEL leads to the following conclusions. Data can be collected in a highly reliable and accurate fashion using meteor burst technology. An analysis of the system performance has shown that the required 90 percent of all sites can be expected to respond to scheduled polls. Diurnal fluctuation of the data is generally less than or equal to 0.3 percent of the sensor's full scale capacity. Data transmitted by SNOTEL can be expected to vary no more than 10 percent from the actual value.

Data from the SNOTEL meteor burst system are used operationally in a variety of ways. Snowmelt runoff in Western U.S. streams is routinely forecast, irrigation applications are being scheduled, and critical water supply shortages have been evaluated and contingency plans formulated. The Soil Conservation Service is reducing its traditional manually measured snow course networks and placing increased reliance on SNOTEL data.

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